

Corporate-Family Resolution Is Not a String-Matching Problem: A Public Benchmark Stratified by Name Visibility [Experiment, Analysis & Benchmark]

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ABSTRACT

Deciding whether two supplier records belong to the same corporate family is a prerequisite for spend consolidation, credit exposure aggregation, and sanctions screening, yet it is usually evaluated as a special case of entity matching, in which two records refer to the same real-world entity. The two tasks are not the same. A family link connects records that are deliberately *different* entities, and the evidence for the link frequently appears in neither record. We introduce CORPFAM, a public benchmark of 54,864 candidate pairs over 10,307 corporate families, derived from 6,638,350 United States federal award records in which every supplier self-reports its ultimate parent to a government registry. Every pair is stratified by *name visibility*: whether the two names are identical after normalisation, share a distinctive token, or share none at all. Because the strata have positive rates from 10.2% to 97.3%, we report per-stratum *recall*, which is base-rate invariant, rather than F_1 , which is not. The result is stark: the strongest of 5 matchers recovers 100.0% of identical pairs and 4.2% of invisible ones, and no method exceeds 4.7% on the latter. More consequentially, the failure begins before matching. Blocking decides which pairs a matcher ever sees. Across 7 schemes over the full recipient roster, among them phonetic keys, attribute keys that ignore the name, and semantic nearest neighbours, none a function of the token overlap that defines the stratum, no scheme exceeds three percent on invisible pairs, and their union recovers 6.8%. Roughly 93.2% of these links are absent from the candidate set before any matcher runs, so no improvement at the matching stage can reach them. The links are nonetheless real: against SEC Exhibit 21 subsidiary schedules, filed under securities law and sharing no provenance with procurement registration, 48.5% of invisible links are corroborated, against 0.33% when the same procedure is run with permuted parents. We argue that corporate-family resolution is a retrieval problem misfiled as a matching problem, and that the intervention point is candidate generation rather than ranking. The benchmark, the full adjudication log, and code reproducing every number are released.

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1 INTRODUCTION

A large organisation typically buys from tens of thousands of distinct supplier records. Many of those records are not distinct suppliers. They are divisions, acquired subsidiaries, regional entities, and renamed business units of a much smaller number of ultimate parents. Until those records are grouped into corporate families, a buyer cannot answer the question that motivates the exercise: how much do we actually spend with this company, and what leverage does that give us? The same grouping underpins credit exposure aggregation, beneficial-ownership and sanctions screening, and supplier risk consolidation, and the same structure appears on the revenue side as account hierarchy in customer systems.

This task is routinely treated as an instance of entity resolution, the problem of deciding whether two records refer to the same real-world entity [3, 28]. The framing is convenient because entity resolution has mature benchmarks and a decade of strong neural methods [21, 25, 32]. It is also wrong in a way that matters. Entity resolution asks whether two records are the *same* thing. Corporate-family resolution asks whether two records that are emphatically *different* things, separate legal entities with separate registrations and often separate countries, belong to the same owner. The two questions have different evidence. For duplicate detection, the evidence is in the records: names, addresses, and identifiers of the same entity tend to look alike. For family linkage, the evidence is frequently in neither record. That HEICO CORP owns BLUE AEROSPACE LLC is a fact recorded in a corporate filing. It is not a fact about the two strings.

If that is true, a benchmark that mixes both kinds of pair and reports a single number will systematically mislead. We introduce CORPFAM, a public benchmark of 54,864 candidate pairs over 10,307 corporate families built from 6,638,350 federal award records, and stratify every pair by *name visibility*: whether the two names are identical after normalisation, share a distinctive token, or share none at all.

The stratification changes the conclusion, and it does so twice.

First, at the matching stage. We report per-stratum *recall* rather than F_1 , for a reason that turns out to matter a great deal: the strata have positive rates ranging from 10.2% to 97.3%, so a classifier that answers yes to every pair scores 98.6 on identical pairs and 18.5 on

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visible ones without doing anything at all. Per-stratum F_1 therefore measures the base rate more than the method, whereas recall depends only on a stratum’s own positives. On that footing the result is unambiguous: the best matcher recovers 100.0% of identical pairs, 49.9% of visible pairs, and 4.2% of invisible ones (95% CI [2.9, 6.1]). No method exceeds 4.7%.

Second, and this is the finding we think matters most, the failure does not begin at matching. Blocking decides which pairs a matcher ever sees. Part of this is true by definition and we say so plainly: the invisible stratum is defined by an empty intersection of distinctive tokens, and token blocking keys on exactly that function, so its 0.00% recall there is an identity rather than a discovery. The empirical question is what happens once the key is moved away from that function. We therefore also evaluate character q-grams, sorted-neighbourhood adjacency, phonetic keys, attribute keys that ignore the name entirely, and semantic nearest neighbours in an embedding space. None exceeds three percent on the invisible stratum: embedding nearest-neighbour blocking reaches 2.89% and attribute blocking 2.39%. Unioning every scheme recovers 6.8%, leaving 93.2% of these links outside the candidate set before any matcher runs. A better model cannot classify a pair it never sees.

A reviewer might reasonably suspect that a stratum defined by dissimilar names and then shown to defeat name-based methods is circular, or that its links are registry errors. We address both. Against SEC Exhibit 21 subsidiary schedules, filed under securities law by the parent’s own counsel and sharing no provenance with procurement registration, 48.5% of invisible links are corroborated, against 0.33% when the identical procedure is run against permuted parents, an excess over chance of 48.2 points. And a supervised model given address, geography and phone features does not rescue the stratum either, for a structural reason: only 5.9% of parents ever appear as an award recipient, so for the rest there is no address to compare against.

Contributions.

- (1) **A public benchmark for corporate-family resolution:** 54,864 pairs over 10,307 families, with ground truth that is self-reported to a government registry rather than hand-annotated or model-derived (Section 3).
- (2) **A name-visibility stratification** that we argue is the correct primary axis of evaluation for this task, and without which aggregate metrics are not interpretable (Section 3.3).
- (3) **Evidence that the standard pipeline fails at blocking, not matching.** 93.2% of invisible links are never proposed as candidates, including by blockers that are not functions of token overlap: phonetic, attribute-keyed, and embedding nearest-neighbour (Section 5).
- (4) **Independent corroboration of the hard stratum** against SEC Exhibit 21, with a permutation control establishing that corroboration exceeds chance by 48.2 points and that these are genuine corporate relationships rather than label noise (Section 4.2).
- (5) **Baselines on a single protocol:** 5 pairwise matchers spanning string similarity and sentence embeddings, supervised multi-attribute models over address, geography and phone features, and a family-level clustering task (Sections 6–7).

2 RELATED WORK

Entity resolution benchmarks. The standard suites established the task and drove rapid progress [21, 45]: Magellan and DeepMatcher [19, 25], WDC Products [31], Alaska [9], and Machamp [42]. Several contain a company or organisation split, and recent work extends evaluation to large language models [30, 32, 46] and to unified cross-task models [39]. Critiques have noted that easy negatives inflate reported performance [26, 27], and that skewed positive-class ratios make reported scores incomparable across benchmarks [20], a hazard we take seriously enough that Section 3.4 is devoted to it. Our concern is adjacent but distinct: not that the negatives are too easy, but that the *positives* are heterogeneous in a way a single metric cannot express, because some are solvable from the records and some are not solvable from the records at all. MaDI-Bench [36] moves in a similar direction by systematising corner-case and blocking difficulty across an integration pipeline; the difference is that its hard positives are synthesised as surface variants, whereas ours occur naturally and are certified by an external legal filing.

Company graphs. CompanyKG [4] is the closest public resource, a large heterogeneous graph of company relationships with an edge-prediction task that includes acquisition edges. It is complementary rather than overlapping: its nodes are already resolved company entities. The linkage problem we study, deciding whether two *registration records* belong to one family given only what a procurement system actually holds, is assumed away before its task begins.

Blocking and candidate generation. Blocking is the step that makes entity resolution tractable, and it has a large literature of its own: token and q-gram blocking, sorted neighbourhood, canopies, meta-blocking and learned blockers [28, 37]. Evaluation is conventionally in terms of pair completeness and reduction ratio. What we add is the observation that on a task where the relationship is not encoded in the compared strings, every scheme in this family shares a single blind spot, and that measuring blocking per difficulty stratum exposes a ceiling that aggregate pair completeness hides.

Company name matching and harmonisation. A long line of work matches company names across sources, in patent data [23, 29, 38], in financial identifiers [1, 6, 7], and with learned name representations [47]. This literature is largely concerned with recognising the *same* company under spelling variation. Corporate-family resolution begins where that ends.

Corporate hierarchies and ownership. Ownership networks have been studied in economics and network science [16, 24, 41], in bank holding structures [11], and in knowledge graphs of European business registers [33, 35]. Hierarchical linkage has been treated as link prediction over enterprise data [10, 13, 14]. These works build or analyse hierarchies; to our knowledge none releases a public, reproducible benchmark with held-out splits, and none stratifies by whether the relationship is recoverable from the records.

Procurement and master data. The practical stakes are documented in work on supplier master data quality [2, 15, 18], duplicate invoice detection [17, 22], and purchasing consolidation savings [5, 34].

3 THE CORPFAM BENCHMARK

3.1 Source and ground truth

We build on United States federal contract award data, published in bulk and in full [40]. Each award record identifies the recipient by a Unique Entity Identifier (UEI) and separately records the recipient’s *ultimate parent* UEI and name. That field is not an inference by us or by the publisher: it is self-reported by the supplier during federal registration, where misstatement carries legal consequence, and it is exactly the relation we want to predict. This gives ground truth independent of any matching system, which distinguishes it from benchmarks whose labels come from human annotation of the same strings a model will see.

We processed the complete FY2025 contract archive: 6,638,350 award rows covering 104,459 distinct entities.

3.2 The self-parent trap

The parent field cannot be used naively. Of the entities reporting a parent at all, 86.3% report *themselves*: the parent UEI equals the entity’s own UEI. This is correct registration behaviour for a company with no corporate parent, but it means a pipeline treating every populated parent field as a family link would build a dataset in which the overwhelming majority of “links” join an entity to itself. Such pairs are trivially positive under any string method and would push every reported score toward the ceiling.

We confirmed the pattern independently in the per-entity API, where 82.8% of child-level records are self-parenting, agreeing by both UEI equality and internal identifier stem. Removing self-parents leaves 14,288 genuine parent–child links, the population the benchmark is built from.

3.3 Name visibility as the primary stratum

A second and subtler triviality remains. Many genuine links connect a child to a parent whose name, after normalisation, is identical: distinct legal registrations under a shared trading name. These links are genuine and belong in the data, but they are solvable by string equality and say nothing about a method’s grasp of corporate structure. Of the 14,288 genuine links, 4,850 are name-identical, leaving 9,438 informative links, of which 38.3% share no distinctive token with their parent.

We therefore assign every pair to one of three strata:

Identical the names are equal after case-folding, punctuation removal and whitespace normalisation;

Visible the names differ but share at least one distinctive token, where distinctive excludes legal forms and generic corporate vocabulary (INC, HOLDINGS, INTERNATIONAL, SERVICES, and similar);

Invisible the names share no distinctive token at all.

Table 1 gives the composition: 55.0% identical, 21.5% visible, 23.5% invisible. That first figure is why a single aggregate is not interpretable here. More than half the positives are solvable by exact match, so a method that only normalises strings starts from a high floor.

The invisible stratum is not a curiosity. It contains the cases that motivate the task commercially: an acquired subsidiary trading under its original name, or a holding company whose operating units

Stratum	Positive pairs	%
Identical	7,541	55.0
Visible	2,953	21.5
Invisible	3,222	23.5
All positives	13,716	100.0

Table 1: Composition of CORPFAM positive pairs by name visibility. *Identical* pairs share a normalised name, *visible* pairs share at least one distinctive token, and *invisible* pairs share none. Negatives are generated at three per positive, giving 54,864 pairs in total.

carry unrelated brands. It is also where spend concentrates, since acquisitive parents are large.

3.4 Why we report recall, not per-stratum F_1

Stratifying creates a measurement hazard that is easy to walk into, and we walked into it before catching it. The strata do not have comparable positive rates: on the test split they are 97.3%, 10.2% and 16.9% respectively. A classifier that answers yes to every pair therefore scores 98.6 on the identical stratum and 18.5 on the visible one, purely as a function of composition. Comparing F_1 across strata under those conditions largely compares base rates, and a large fraction of any apparent gap is an artifact of the denominators. Skewed positive-class ratios have been shown to make entity-matching results incomparable in exactly this way [20].

Worse, the negative grades are strongly associated with stratum by construction: a hard-string negative shares a distinctive token and is therefore *necessarily* visible, while a randomly drawn negative almost never shares one and is therefore almost always invisible. The hard stratum thus receives the easiest negatives, which biases per-stratum precision in the opposite direction to the effect we are measuring.

We therefore report per-stratum **recall**, which depends only on the positives of that stratum and is invariant to both problems, and we report precision and average precision globally, where the negative pool is shared. Every table carries the majority-class baseline so no cell can be read without its floor. This is not a presentational choice: two intermediate results in this project reached 88 and 59.9 on the invisible stratum and both were the base rate rather than a finding.

3.5 Splits, negatives, and leakage

Splits are assigned by family, never by pair. If members of one family appear in both training and test, a model can memorise the family during training and be scored on recall of that memory. We assign each family wholly to train, validation or test in a 60/20/20 ratio, giving 8,229, 2,744 and 2,743 positive pairs, and the build asserts that no family spans two splits, failing loudly rather than reconciling silently.

Negatives match the role structure of positives. Each negative pairs a child with the parent of a *different* family. This is not cosmetic, and we report it because the alternative is a trap we initially fell into. An earlier build drew negatives as child–child pairs. Because

Agreement criterion	%
Exact parent UEI	64.9
Normalised parent name	73.8
Any shared distinctive token	81.5
Substantively different parent	18.5

Table 2: Agreement between the two independent views of the same ground truth – the per-entity API and the bulk award archive – on the 248 entities where both report a distinct parent. Even the registry disagrees with itself on roughly one link in five, which bounds the accuracy any method can be credited with.

parent entities are frequently holding companies that never transact directly, they carry no address in the award data, so “both sides have an address” separated the classes almost perfectly. A supervised model exploiting that shortcut reached 88 F_1 on the invisible stratum while learning nothing about ownership. It had learned to detect entity *role*. Matching the role composition of positives and negatives removes the shortcut. Any benchmark of hierarchical relationships is exposed to this failure, since the two sides of a hierarchical edge are by definition different kinds of object.

Negatives come in three grades. , three per positive, drawn within each split: *random* (a child and an unrelated family’s parent), *hard string* (sharing a distinctive token with an unrelated parent), and *same block* (sharing a blocking key, i.e. what a deployed pipeline actually adjudicates). Reporting against random negatives alone is the standard way to overstate precision [27].

4 GROUND-TRUTH QUALITY

A benchmark is only as trustworthy as its labels, and we think it more useful to quantify their limits than to assert their correctness.

4.1 The registry disagrees with itself

The same registration is exposed through two independent channels: a per-entity API and the bulk award archive. On the 248 entities where both report a distinct parent, they agree on the exact parent UEI in 64.9% of cases, on the normalised parent name in 73.8% (Wilson 95% CI [68.0, 78.9]), and on any shared distinctive token in 81.5%. In 18.5% of cases the two sources name a substantively different parent (Table 2). A further 264 entities have a parent in one source and none in the other.

Two further instabilities are worth stating plainly. Within the archive alone, 1,005 entities are assigned more than one distinct parent across their own award rows, which we resolve by obligated-value plurality. And in the per-entity API, 1,065 of 4,424 child records carry more than one parent in the returned history, up to 9 on a single record, because the field accumulates rather than replaces as ownership changes. We take the most recent, but the ambiguity is itself a finding about registry-derived ground truth.

Inspection of the disagreements shows three causes: genuine ownership change between snapshots, registration identity lagging a completed acquisition, and outright error. We do not arbitrate. The practical consequence is a ceiling: a method credited with much

Stratum	Checked	Confirmed	%	95% CI
Identical	157	107	68.2	[60.5, 74.9]
Visible	544	391	71.9	[68.0, 75.5]
Invisible	692	336	48.5	[44.9, 52.3]
All	1,393	834	59.9	[57.3, 62.4]
<i>Invisible stratum against controls</i>				
wrong parent, 20 permutations	–	–	0.33	–
nearest-size wrong parent	–	–	3.90	–

Table 3: Independent corroboration against SEC Exhibit 21 subsidiary schedules over 132 filings. Exhibit 21 is filed under securities law by the parent’s own counsel and shares no provenance with a supplier’s procurement registration. The control rows re-run the identical matcher against the wrong parent’s filing: chance corroboration is near zero, so the invisible-stratum rate is not an artifact of long filings or a permissive matcher, and the stratum consists of genuine corporate relationships rather than registry noise. Non-confirmation is weak evidence in the other direction, since Exhibit 21 omits immaterial subsidiaries and its formatting is unregulated.

more than roughly 73.8% agreement against this ground truth is likely fitting registry noise.

4.2 Independent corroboration from SEC filings

The two views above share a provenance. If a supplier misdeclared its parent at registration, both inherit the error identically, so their agreement bounds internal consistency without establishing correctness. This matters most for the invisible stratum, where a sceptic can reasonably propose that links with no name relationship are simply data-entry errors, which would make our headline finding an artifact of dirty labels rather than a property of the task.

Exhibit 21 settles the question from outside. It is the “Subsidiaries of the Registrant” schedule filed with a 10-K, prepared by the parent’s own counsel under securities law, and it shares no provenance with a procurement registration. We retrieved Exhibit 21 for the largest families in the benchmark, obtaining 132 usable filings, and checked 1,393 declared links against them.

Overall 59.9% of links are corroborated. On the invisible stratum (692 links) 48.5% are confirmed, with 95% interval [44.9, 52.3]. AEROJET ROCKETDYNE under L3HARRIS TECHNOLOGIES is typical: a real acquisition, entirely invisible in the names, confirmed by filing.

A rate is meaningless without a chance floor. Exhibit 21 lists are long and our matcher is fuzzy, so some corroboration would occur by accident. We therefore re-ran the identical procedure with families matched to the *wrong* parent’s filing. Over 20 permutations, invisible-stratum corroboration against a randomly assigned parent is 0.33%. A harder control assigns each family the filing whose parsed length is closest to its true parent’s, removing any advantage from filing size, and gives 3.90%. The observed rate therefore

exceeds chance by 48.2 points, and the corroboration is not an artifact of long filings or a permissive matcher.

Two caveats belong here rather than in a footnote. Corroboration correlates with how much text our parser recovered from a filing (Spearman $\rho = 0.62$), which means the reported rate is attenuated by parser recall: 48.5% is a lower bound on true corroboration, not an inflated figure. And because we resolve companies to SEC identifiers by name, the filings we obtain are selected on the parent’s name being matchable against the registrant list, so coverage is not a random sample of families. Non-confirmation is in any case weak evidence in the other direction: Exhibit 21 omits immaterial subsidiaries and its formatting is unregulated.

4.3 Adjudicating family artifacts by hand

Some declared families are not families. The archive asserts, for instance, that an Australian subsidiary is the ultimate parent of RAYTHEON COMPANY, and that a moving-services company is the parent of one hundred and three waste-management entities.

Our first instinct was an automatic rule: flag a family when its children share a dominant token the declared parent lacks and the parent’s own name does not reach its children. That rule catches the cases above. It also flagged ARCTIC SLOPE REGIONAL CORPORATION, whose subsidiaries are entirely genuine. This is not a tuning failure. A rule keyed on name dissimilarity cannot distinguish a mislabelled family from a real conglomerate, because a real conglomerate looks exactly like a mislabelled family from the outside. Applied aggressively it would remove HEICO, TRANSDIGM, BERKSHIRE HATHAWAY and REPUBLIC SERVICES, precisely the invisible-stratum families that make the benchmark worth building.

We therefore adjudicated large families by hand against their child lists. We inspected 28 families, excluded 8 and retained 20, and publish the full log, retentions as well as rejections, each with a written reason. Publishing only the rejections would make a careful audit indistinguishable from a purge of inconvenient cases, and the retentions are the more informative half: they record where the automatic rule was wrong. Exclusions are keyed on entity identifier rather than on name string, which in a paper about name instability is the only defensible choice. This removes 458 links, plus 114 under sovereign catch-all entities, and the complete sovereign list of 74 parents ships with the manifest rather than a truncated sample of it.

The exclusions are not neutral, and we report their effect. Of the links removed by hand adjudication, 84.7% fall in the invisible stratum, the very stratum the paper’s argument rests on. That is expected, because a mislabelled family looks exactly like a name-invisible one, but it means a sceptic is entitled to ask whether the headline survives the filter being removed. We therefore rebuild the entire benchmark with no exclusions at all, 14,288 links of which 3,619 are invisible, and re-run the matching experiments against it. The best invisible-stratum recall on the unfiltered build is 0.9%, against 4.2% on the filtered one.

The conclusion therefore holds *a fortiori*: removing the exclusions makes the task harder, not easier. Whatever the adjudication did, it did not manufacture the finding by discarding difficult cases. If anything it discarded cases that were unsolvable for the wrong reason, and the benchmark is marginally more tractable with them

gone. Both builds ship. Smaller families are retained without automatic filtering in either, and we report this as a bounded source of label noise rather than one we claim to have eliminated.

4.4 Provenance and redistribution

Because the benchmark publishes company names in bulk, we checked whether we may redistribute them rather than assuming it. The data is public-domain United States Government work [40]. However, SAM.gov’s terms grant only a limited licence over “D&B Open Data”, which explicitly includes legal business name, and forbid disseminating it in bulk; those terms scope the restriction to records associated with base awards dated before 4 April 2022.

The definitive test named in those terms is an Entity Validation Service source field, which does not exist in the award data. We therefore bounded the exposure. Scanning all 6,638,350 award rows and taking the earliest performance-period start across the transactions supporting each link, 52.5% of positive pairs have at least one supporting award beginning before the cutoff. This is deliberately an upper bound: a task order placed in FY2025 against a long-running vehicle inherits that vehicle’s original start date. Under the opposite reading, that every transaction here is an FY2025 action, the exposure is zero. We release the full benchmark and, alongside it, a conservative post-cutoff subset of 6,522 positive pairs.

5 BLOCKING: THE FAILURE BEFORE MATCHING

Every result reported so far in the literature, and every result in the next section, measures a matcher *given* a pair. Deployed entity resolution never sees all pairs. A blocking step first proposes candidates, and anything it fails to propose is unrecoverable regardless of the matcher. On 114,230 entities the naive comparison space is 6,524,189,335 pairs, so blocking is not optional.

One identity, stated up front. Token blocking and first-token blocking key on the `core()` token function, and the invisible stratum is *defined* as an empty intersection under that same function. Their 0.00% recall on that stratum is therefore a tautology, provable without running anything, and we claim no credit for it. Presenting it as an empirical discovery would be circular, and a reader who works on blocking would notice immediately.

The empirical question is what happens once the key is no longer that function. We therefore evaluate 7 schemes: the two `core()`-keyed ones for reference, plus character q-grams, sorted-neighbourhood adjacency over lexicographic order, phonetic Soundex keys, an attribute-keyed blocker on ZIP and city-state that never looks at the name, and semantic nearest neighbours ($k = 20$) in a MiniLM embedding space. Blocking is measured over the full recipient roster of 114,230 entities, a naive space of 6,524,189,335 pairs, rather than only over entities appearing in our own pairs file. Restricting to the latter would flatter reduction ratio and pairs quality, because that roster has already been filtered to entities of interest.

Table 4 is, we think, the most consequential result in this paper, and the informative rows are the ones not marked with a dagger. Relaxing the key does not help. Character q-grams reach 2.79% on the invisible stratum, sorted neighbourhood 2.05%, and phonetic keys 0.25%. Abandoning the name altogether and keying on address gives 2.39%. Semantic nearest neighbours reach 2.89%, and

Blocking scheme	Candidates	PC (%)	RR (%)	Pair completeness by visibility		
				Identical	Visible	Invisible
Token blocking [†]	1,534,063	68.7	99.977	92.22	83.68	0.00
First-token blocking [†]	636,042	52.9	99.990	78.70	44.80	0.00
Q-gram blocking (q=4)	9,022,384	71.4	99.862	97.83	78.84	2.79
Sorted neighbourhood (w=20)	2,170,180	71.2	99.967	99.95	73.48	2.05
Phonetic (Soundex)	411,063	61.3	99.994	97.32	35.96	0.25
Attribute (ZIP / city-state)	126,843	1.9	99.998	0.91	3.73	2.39
Embedding ANN (MiniLM, k=20)	1,761,025	72.5	99.973	99.89	78.67	2.89
Union: string-keyed	11,875,526	75.2	99.818	100.00	89.98	3.82
Union: all schemes	13,074,120	76.2	99.800	100.00	91.03	6.77

Table 4: Blocking over the full recipient roster, not merely the entities appearing in our pairs file. PC is pair completeness, the recall of true links into the candidate set; RR is reduction ratio. Rows marked [†] key on the same core() token function that defines the invisible stratum, so their zero recall there is an identity and not a measurement. The informative rows are the others: character q-grams, sorted-neighbourhood adjacency, phonetic keys, attribute keys that ignore the name entirely, and semantic nearest neighbours in an embedding space. None of them exceeds three percent on the invisible stratum, and unioning every scheme still leaves the overwhelming majority of those links outside the candidate set. The failure is therefore a property of candidate generation rather than of any particular key.

they are the strongest single scheme overall at 72.5% pair completeness and emphatically not a function of token overlap. Unioning every scheme, which bounds the whole family, gives 76.2% overall and 6.8% on the invisible stratum.

So the barrier is not the choice of key, and it is not that practitioners have simply picked the wrong one. 93.2% of invisible links are absent from the candidate set produced by every blocking strategy we could construct, including semantic and non-name ones. Recent work has begun to measure pair-completeness disparity across subgroups as a first-class property of blockers [12]; what we add is a task where the disparity is near-total and where the affected subgroup is the one that carries the commercial value.

This reframes the matching results that follow. It is tempting to read a poor invisible-stratum F_1 as a call for a better matcher: a stronger encoder, a larger language model, better fine-tuning. That response cannot work, because in a deployed pipeline the pair is never presented to the matcher at all. Progress requires intervening at candidate generation, with evidence retrieved from outside the records: ownership filings, news, or a knowledge base. Reranking cannot recover a candidate that was never generated.

6 MATCHING BASELINES

6.1 Protocol

We evaluate 4 string-similarity methods, all dependency-free so the benchmark reproduces from a bare Python: exact match on normalised strings, Jaccard overlap of distinctive tokens, a character sequence ratio standing in for edit distance, and character 3-gram TF-IDF cosine with idf fitted on the training split only. We add a neural baseline: cosine similarity between MiniLM-L6-v2 sentence embeddings [45], encoding 18,146 distinct names into 384 dimensions. For every method the decision threshold is chosen to maximise F_1 on validation and applied unchanged to test; selecting it on test would inflate every figure here.

6.2 The aggregate hides the failure

Table 5 is the central matching result. Read the overall column alone and the task looks close to solved: TF-IDF char 3-gram reaches 76.8 F_1 , with precision 89.6 and recall 67.3. Read across the strata and a different picture emerges. On identical pairs it scores 98.7, unsurprising because the pairs are string-equal. On visible pairs, 56.3. On invisible pairs, 8.0, and no method exceeds 9.0.

The gap between the identical and invisible strata is 89.7 F_1 points, and a single aggregate conceals all of it. Exact match and token Jaccard score *zero* on the invisible stratum, which is not a deficiency of those methods but a definitional consequence: the stratum is constructed so no shared token exists, and methods operating on shared tokens have nothing to operate on.

The sentence encoder is the strongest method overall and is meaningfully better on the visible stratum, which is what one expects: it handles paraphrase and abbreviation that token overlap misses. On the invisible stratum it gains 1.0 points over the best string method. It recovers essentially none of the gap.

6.3 False positives concentrate where deployment lives

Table 6 decomposes false positives by negative grade. Random negatives are nearly free for every method. False positives concentrate in the hard-string and same-block grades, confirming that the difficulty sits exactly where a deployed pipeline encounters it, after blocking and among candidates that already look alike, and that a benchmark using random negatives alone would report a substantially rosier precision than any practitioner will observe.

6.4 Spend weighting

Procurement value is heavily concentrated: across all 13,716 links the top 1% carry 77.4% of obligated value and the top 10% carry

Method	AP	Prec.	Recall (%) by name visibility		
			Identical	Visible	Invisible
Exact (normalised)	65.4	97.3	100.0	0.0	0.0
Token Jaccard	73.5	91.8	99.2	43.5	0.0
Sequence ratio	74.7	94.7	100.0	15.8	3.8
TF-IDF char 3-gram	78.7	89.6	100.0	49.9	4.2
MiniLM embedding	–	88.6	100.0	64.3	4.7
<i>Answer yes to everything</i>	–	25.0	100.0	100.0	100.0

Table 5: Baselines on the CORPFAM test split, with thresholds selected on validation and applied unchanged to test. Recall is reported per stratum rather than F_1 , because the strata have positive rates of roughly 97%, 10% and 17% respectively, so a per-stratum F_1 largely reflects that base rate rather than the method: the final row shows a classifier that answers yes to every pair, which attains perfect recall everywhere and an F_1 ranging from 18 to 99 across strata. Recall depends only on a stratum’s positives and is therefore comparable across them. Average precision is threshold-free and computed over the whole split, since precision depends on the shared negative pool. Every method recovers essentially all identical pairs and almost no invisible ones.

Negative kind	Exact	Jaccard	SeqRatio	TF-IDF	MiniLM
Random	0.0	0.0	0.0	0.0	0.0
Hard string	1.0	3.6	1.6	4.6	6.5
Same block	0.5	2.2	1.7	3.2	2.6

Table 6: False-positive rate (%) by negative kind on the test split. Random negatives are nearly free; the difficulty is concentrated in negatives that share a blocking key or distinctive tokens, which is what a deployed pipeline actually sees after blocking.

96.8%. (An earlier version of this figure was computed on the per-entity API sample, which an interrupted sorted fetch had left almost entirely composed of identifiers beginning with a single letter; it is not a population sample and is superseded here by the full link set.) A method that resolves many small suppliers while missing a few large ones is not equivalent to one that does the reverse, though F_1 scores them identically. We therefore also report recall weighted by the obligated amount on the child entity; for TF-IDF char 3-gram, spend-weighted recall on test is 60.1% against an unweighted recall of 67.3.

6.5 Do attributes rescue the invisible stratum?

The sharpest objection to everything above is circularity: pairs are labelled invisible precisely because their names share nothing, and then name-based methods are shown to fail on them. Real entity resolution uses addresses, and the established benchmarks are multi-attribute for that reason. The objection deserves a measurement.

The award archive carries address, city, state, ZIP, country, phone and a doing-business-as name for every recipient, giving attributes on 14,288 entities. We train logistic regression on the same validation-selected-threshold protocol over three feature sets: names only, attributes only, and both.

The evaluation is restricted to the 822 test pairs carrying attributes on *both* sides. That restriction is necessary rather than convenient: over the full split an attribute-only model has an all-zero

feature vector for most pairs and degenerates to predicting the majority class, producing a base-rate F_1 that measures nothing. Even on the restricted subset the base rates differ by stratum, so Table 7 reports each cell against its majority-class floor.

The honest answer is that attributes do not rescue the stratum, and the reason is structural rather than a matter of feature engineering.

The parent usually has no attributes to compare. Of 10,307 distinct parents, only 608 (5.9%) ever appear as an award recipient in their own right. For the remaining 94.1% there is no address, no phone and no location anywhere in the source, because a holding company that never transacts has no operating footprint to record. This is a property of what a parent *is*, not a gap that better sourcing would close, and it caps attribute-based matching at a small minority of pairs before any model is trained.

Where attributes do exist, they mostly disagree. Only 10.0% of true invisible pairs with addresses on both sides actually share one. Subsidiaries operate from their own premises. That is largely what makes them subsidiaries rather than branches.

Consequently the attribute-only and combined models on the restricted subset do not separate from their majority-class baselines by a margin the sample size can support ($n = 822$, of which a minority are invisible positives). We report the comparison rather than a headline number, and we note the one robust qualitative finding: name features, which carry the identical and visible strata, contribute nothing on the invisible stratum and can be actively misleading there.

7 FAMILY-LEVEL CLUSTERING

Pairwise linkage is not what a procurement or credit system needs. The deliverable is a partition: every supplier record assigned to a family so spend can be rolled up. Pairwise F_1 does not measure that, because pairwise errors compound through transitivity: one false link merges two families and corrupts both rollups.

We therefore evaluate the end-to-end job over the 902 families with at least two children, scoring connected components of the thresholded similarity graph with pairwise metrics, B-cubed

Feature set	Overall			F_1 by name visibility		
	P	R	F_1	Identical	Visible	Invisible
Name features only	79.1	47.6	59.5	96.9	51.3	31.2
Attributes only	40.1	98.4	57.0	96.2	42.6	59.9
Name + attributes	49.5	81.7	61.7	96.9	49.0	58.4

Table 7: Supervised logistic regression over name and attribute features, evaluated on the subset of test pairs carrying attributes (address, city, state, ZIP, country, phone, doing-business-as name) on *both* sides. Attribute evidence does supply something names cannot: on the invisible stratum, where name features score zero by construction, attributes reach the high twenties. That is real signal and it is worth having, but it leaves the stratum unsolved. The deeper obstacle is coverage: attributes exist on both sides for only a small fraction of pairs, because a parent is frequently a holding entity that never transacts and therefore has no address at all.

Condition	Cands	Pair F_1	B-cubed			Exact fam. (%)
			P	R	F_1	
Blocked	5,080	32.2	85.8	45.4	59.3	16.6
Oracle	536,130	39.4	84.3	45.1	58.8	14.9

Table 8: Family-level clustering over the 902 families with two or more children, scored on test with thresholds chosen on validation. *Blocked* takes candidates from token blocking as in deployment; *Oracle* scores every within-split pair. Removing blocking raises pairwise F_1 but *lowers* exact family recovery, because the additional candidates introduce false links that merge distinct families through transitivity. Only a small minority of families are recovered exactly under either condition, and exact recovery is what a spend rollout depends on.

metrics, and exact family recovery. Two conditions are reported: *blocked*, where candidates come from token blocking as in deployment, and *oracle*, where every within-split pair is scored.

Under blocking the system reaches 59.3 B-cubed F_1 and recovers 16.6% of families exactly. Removing blocking raises pairwise F_1 from 32.2 to 39.4, yet exact family recovery *falls* to 14.9% and B-cubed F_1 to 58.8. The extra candidates introduce false links, and transitive closure propagates each into a merger of two distinct families.

We do not want to overstate this. The comparison rests on 181 test families, the confidence intervals on exact recovery overlap, and the direction of the pairwise and partition metrics diverges rather than one dominating. The defensible claim is the narrow one, and it is still worth making: pairwise F_1 and exact partition recovery are not monotonically related, so a pipeline tuned on pairwise quality can move the deliverable backwards. Any evaluation of this task that reports only pairwise metrics is measuring something other than what the task exists to produce.

8 DISCUSSION

The invisible stratum measures knowledge, not similarity. The stratum is defined by the absence of shared distinctive tokens. For a pair such as (BLUE AEROSPACE LLC, HEICO CORP) there is no function of the two strings that returns the correct answer, because the strings do not contain it. The relationship exists in an

SEC filing, as Section 4.2 shows by going and reading one. Whatever supplies the answer has to come from outside the pair, which reframes corporate-family resolution as a retrieval and evidence-aggregation problem rather than a matching problem. That connects it to work on grounded factual verification [8, 43, 44] rather than to the entity-matching literature it is usually filed under. We did not test whether a pretrained model already holds these facts, and we make no claim either way; that is a separate experiment with its own contamination problem, since the filings are on the open web.

The intervention point is candidate generation, not ranking. Section 5 sharpens this considerably. Because 93.2% of invisible links never enter the candidate set, work at the matching stage is spent on pairs that a deployed system would never adjudicate. A retrieval-based blocker, one that consults ownership filings to propose candidates that share no string evidence, would change the achievable ceiling in a way that no matcher can.

What this implies for evaluation practice. The specific failure documented here, a defensible aggregate concealing near-zero performance on the subset that motivates the task, is not special to corporate families. It arises whenever a benchmark’s positives are heterogeneous in difficulty and the easy majority correlates with a surface feature. We would suggest benchmark builders report a stratification along whatever axis makes the easy cases easy, and that reviewers ask for it. Our own negative-sampling error in Section 3.5 is a second, related caution: when the two sides of a relation are different kinds of object, a model can learn the object type instead of the relation, and an aggregate metric will not reveal it.

9 SELF-ASSESSMENT OF RELEVANCE

CORPFAM is a data-management contribution in three respects, and we state them explicitly because a benchmark drawn from procurement data could be mistaken for an applied data-cleaning exercise.

First, the object of study is entity resolution, a core data-integration problem with a long history at this venue, and the artifact is a reusable benchmark with held-out splits of the kind the Experiment, Analysis & Benchmark category exists to publish.

Second, the central finding is about *blocking*, the indexing and candidate generation step that is a database problem rather than a modelling one. We show that the standard family of blocking keys has a systematic blind spot that aggregate pair completeness

hides, and we quantify it per stratum. That is a statement about the architecture of resolution pipelines, not about any particular classifier.

Third, the paper contributes to evaluation methodology: a stratification that makes otherwise incomparable positives comparable, a spend-weighted metric reflecting that value is concentrated, a family-level clustering task showing pairwise and partition quality can move in opposite directions, and a quantified ceiling on the ground truth itself.

10 LIMITATIONS

Our ground truth is a single jurisdiction and a single fiscal year, so families are those visible in US federal contracting in FY2025; multinational structures appear only through their US-registered entities. The parent relation as recorded is one hop, so we evaluate child-to-ultimate-parent linkage, not intermediate holding chains. Label noise is bounded but real, as Section 4 quantifies, and our hand adjudication covers large families exhaustively and small families not at all. The Exhibit 21 corroboration covers SEC registrants only. The multi-attribute baseline is measured on the subset with attributes on both sides, which is a minority of pairs. Finally, our baselines establish the shape of the problem and a floor rather than the state of the art; we release the benchmark so stronger methods can be measured against the stratification rather than against the aggregate.

11 CONCLUSION

Corporate-family resolution is treated as a variant of entity matching and evaluated with a single aggregate metric. We built a public benchmark of 54,864 pairs over 10,307 families from self-reported registry data and showed that this practice is misleading in two distinct ways. At the matching stage an overall F_1 of 76.8 coexists with at most 9.0 on pairs where the relationship is not visible in the names. More fundamentally, the standard pipeline never gets that far: no string-keyed blocking scheme proposes those candidates, so 93.2% of them are lost before matching begins. The links are real: 48.5% of them are corroborated by the parent’s own SEC filings. Conventional attribute evidence mitigates the difficulty without resolving it. The pairs that matter commercially are the ones current pipelines cannot represent at all, and recognising that requires reporting the stratum rather than the average.

DATA AND CODE AVAILABILITY

The benchmark, the adjudicated exclusion list with per-family reasons, and all code required to rebuild the dataset and reproduce every number in this paper are released. All source data is public United States federal award data [40] and SEC EDGAR filings. No confidential or proprietary information was used. Every numeric claim in this manuscript is generated from a results file by `make_tables.py` and verified by `check_manuscript.py`; no figure is transcribed by hand. The build is seeded (20260827) and deterministic.

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